



Home MaintenanceServiceRequest Platform
ProjectProposal —Final iTi Node.js

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1. Problem Statement

When an appliance breaks down or a maintenance issue appears at home (washing machine, AC, plumbing, electrical, and similar problems), people struggle to quickly find a trustworthy technician, compare prices, and track the status of the request. There is no organized way to connect the person requesting the service with the right specialist transparently and with a clear record of the work.

2. Delivered Solution

FixIt lets a customer create a service request describing the problem, the required specialty, and an optional photo. Registered technicians in that specialty view pending requests, and can search or filter them by specialty, status, or title. A technician submits an offer with a price and estimated time. The customer accepts one offer, which automatically rejects the remaining offers on that request. The technician marks the job completed once finished, and the customer then rates the technician. An admin account can manage users (ban/unban), view and delete requests, and view and delete reviews.

3. Roles

Role	Responsibilities
Customer	Creates a service request, reviews technician offers, accepts one, rates the technician after completion
Technician	Browses pending requests (with search/filter), submits offers, marks accepted jobs as completed
Admin	Manages users (ban/unban), monitors and deletes requests, monitors and deletes reviews

4. Delivered Features

- JWT authentication for all three roles, with role embedded in the token
- Role-based authorization on every protected route (customer / technician / admin)
- Service request creation with photo upload, validated for file type and size
- Specialty-based matching across 12 specialties (Plumber, Electrician, AC Technician, Carpenter, Painter, Mason, Appliance Technician, Locksmith, Glass & Aluminum Technician, Cleaning Technician, Gardener, Electronics Technician)
- Search and filter on requests by specialty, status, and title
- Offer submission and atomic offer acceptance (race-condition safe)
- Ownership checks on completing a request and on accepting an offer
- Rating system limited to one rating per completed request
- Admin dashboard for user management and review moderation
- Centralized error handling middleware
- Full Swagger documentation and a Postman collection

5. Request Flow (As Built)

- Customer creates a service request (title, description, specialty, location, optional photo).

- Technicians in that specialty see it in their pending list, with search/filter available.
- One or more technicians submit an offer (price + estimated time).
- Customer accepts one offer; every other offer on the same request is automatically rejected.
- Request status moves from pending to accepted.
- The assigned technician marks the request completed once the job is done.
- The customer submits a single rating (stars + comment) for the technician.

6. Tech Stack

Item	Technology
Runtime	Node.js
Framework	Express.js
Database	MongoDB + Mongoose
Auth	JWT Authentication
Validation	express-validator (register endpoint)
File Upload	Multer
Error Handling	Centralized error-handling middleware
Docs	Swagger + Postman Collection
Frontend	Plain HTML / CSS / vanilla JavaScript
Versioning	Git + GitHub, multi-author commit history

7. Architecture

The backend follows a plain layered structure: routes define the URL and which middleware/controller handles it, controllers hold the business logic, models define the MongoDB schemas, and middleware covers authentication, authorization, validation, file upload, and error handling. This keeps every layer single-purpose and easy to explain file by file.

8. Access and Documentation

Item	Location
GitHub repository	https://github.com/Ateff22/FIX-IT
README	README.md in the repository root
Postman collection	FixIt.postman_collection.json in the repository root
Swagger UI	http://localhost:5000/api-docs (while the server is running locally)

9. Status

The project is complete and functioning end to end, covering the bootcamp's minimum technical requirements. It has been tested manually across all three roles.